Operations Research

Linear Programming IV: Revised Simplex

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Linde / MDSI Scholarship Program

Program

Linde / MDSI Master's Scholarship Program

Female students are especially invited to apply

Funding & Duration

- 1000€ monthly for 2 semesters
- funding start: upcoming semester 26W

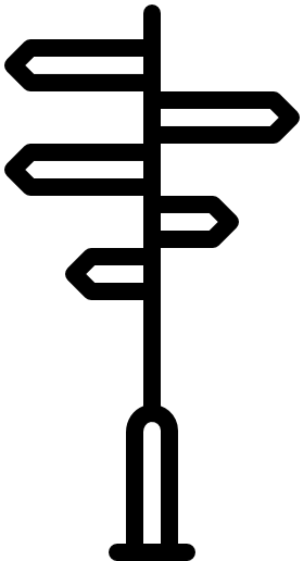
Eligible to apply are students from

- Data Engineering & Analytics
- Mathematics in Data Science

Further details & application material

- https://go.tum.de/804736

Course outline



- **Linear Optimization**
  - Modeling by linear programming and graphical solution
  - Solving linear programs, convexity
  - Simplex algorithm, 2-phase simplex
  - **Simplex algorithm in matrix notation**
  - Sensitivity analysis, re-optimization
  - Duality theory, zero-sum games

- **Integer Linear Optimization**
  - Modeling integer programming problems (IPs)
  - Solving IPs: branch & bound, cutting planes
  - Advanced solution methods: column generation, approximation algorithms
  - Fast solvable integer problems: total unimodularity, matroids
  - Network flow problems
  - Traveling salesperson problem

- **Nonlinear Optimization**
  - Optimization of multidimensional functions
  - Convex optimization

Today you will learn ...

- how to formulate the Simplex algorithm in matrix notation and how to run the **revised Simplex algorithm**.
  - This will be the central notation we use in the next lectures.

- Apply linear programming for **classification problems**.

- Understand **vector optimization** and how you deal with conflicting objectives.

If you haven‘t done so, please exercise to understand the Simplex algorithm.

LP: what we learned so far

- Graphical method → Corner point solutions
- Corner point solutions of the standard form → basic solutions and
  adjacent basic solutions
- Algebraic Simplex method

Hereafter:

- Simplex method in matrix notation (aka. revised Simplex).

Canonical form, basic variable (recap)

Canonical form:

$$
\begin{aligned}
\max\ & c^T x \\
\text{s.t.}\ & Ax = b \\
& x \geq \vec{0}
\end{aligned}
$$

Each equation contains a variable with coefficient 1, which has a coefficient of 0 in all other equations (for each constraint there is an associated basic variable).

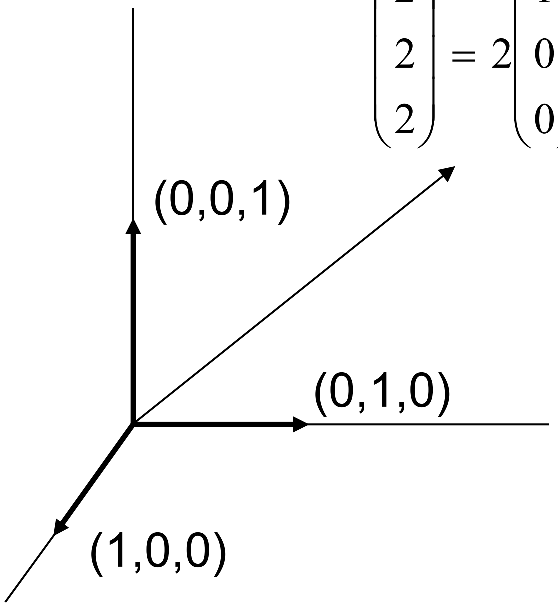
Basic variable: x_i is called basic variable for constraint j, if

$$c_i = 0, b_j \geq 0 \text{ and } a_i = e_j = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \leftarrow \text{j-th row.}$$

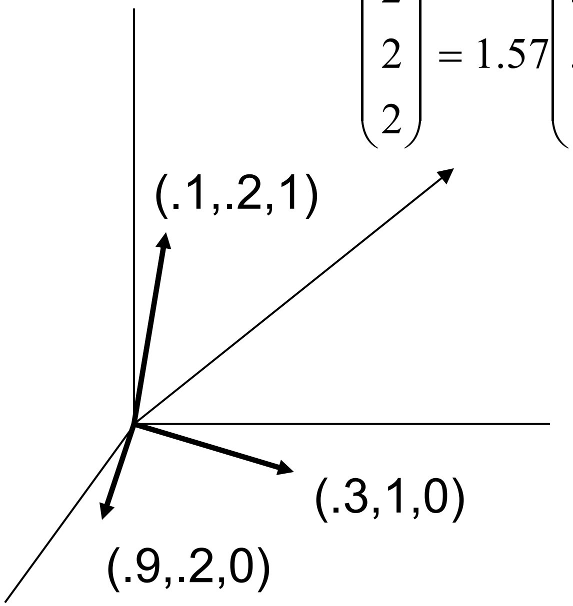
Linear independence and basis

A **basis** is a set of linearly independent vectors which span the space.
With three dimensions we need tree basis vectors of dimension 3.
A basis is a minimal set of spanning vectors (not necessarily orthogonal).

$$\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



$$\begin{pmatrix} 2 \\ 2 \\ 2 \end{pmatrix} = 1.57\begin{pmatrix} .9 \\ .2 \\ 0 \end{pmatrix} + 1.29\begin{pmatrix} .3 \\ 1 \\ 0 \end{pmatrix} + 2\begin{pmatrix} .1 \\ .2 \\ 1 \end{pmatrix}$$



Linear dependency

A set of columns (or rows) of a matrix is **linearly independent** if no column (row) can be expressed as a linear combination of other columns (rows).

The **rank** of a matrix is defined as the number of linearly independent rows (or columns) of a matrix.

Nonsingular (regular) matrix

- Any square matrix that has no linear dependencies between the rows (or columns). The determinant is non-zero.

Singular matrix

- There exists at least one linear dependency between the rows (or columns) of the square matrix. The determinant is zero.

Finding an initial basis (recap)

If the LP is given in canonical form, we can easily find a BFS.

Example: LP in normal form with positive right side: introduce slack variables and select them as basic variables (the zero point is the first BFS).

This is equivalent to the origin being part of the feasible region!

If this is not the case, we have to calculate a first BFS.

Two options:

- 2-phase method
- Big-M method

Basic solutions

- How do we find the basic solution when the basis is known?

- Example: basis $\{x_1, x_3\}$ (e.g. given by the Big-M method)

$$
\begin{aligned}
\max\ & -x_1 - 2x_2 \\
s.t.\ & 2x_1 + x_2 - 4x_3 = -4 \\
& x_1 + 2x_2 + 2x_3 = 4 \\
& x_1, x_2, x_3 \geq 0
\end{aligned}
$$

x_2 set to 0 and
→ Solve the 2×2 system: $2x_1 - 4x_3 = -4$
$x_1 + 2x_3 = 4$

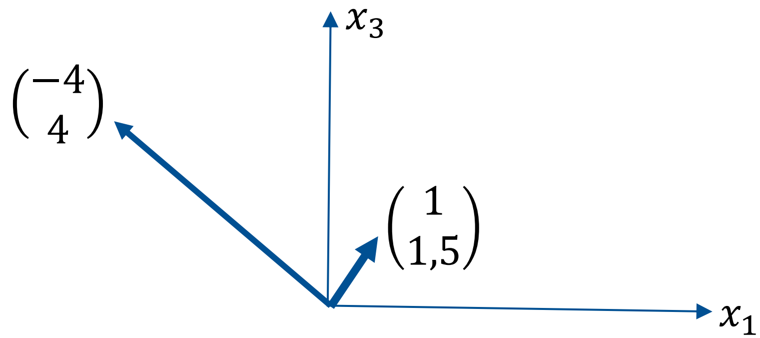
Note: The second constraint has been changed (compared to the last example). Thus, the result is not yet known.

Linear transformations via matrices

$Ax = b$ and $x = A^{-1}b$ describe linear transformations from one vector x to another vector b and back.

$$\begin{bmatrix} 2 & -4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 & -4 \\ 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$



The identity matrix $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ does nothing to a vector.

If the determinant $\det(A) = 0$, then the transformation squishes space into a zero area region. Such matrices do not have an inverse and they do not have full rank.
Matrix A in the example has **full rank** of 2 and the determinant $\det(A) = 8$.

Null space of a matrix

Definition: The null space of an $m \times n$ matrix A, written as Nul A, is the set of all solutions of the **homogeneous equation** $Ax = 0$.

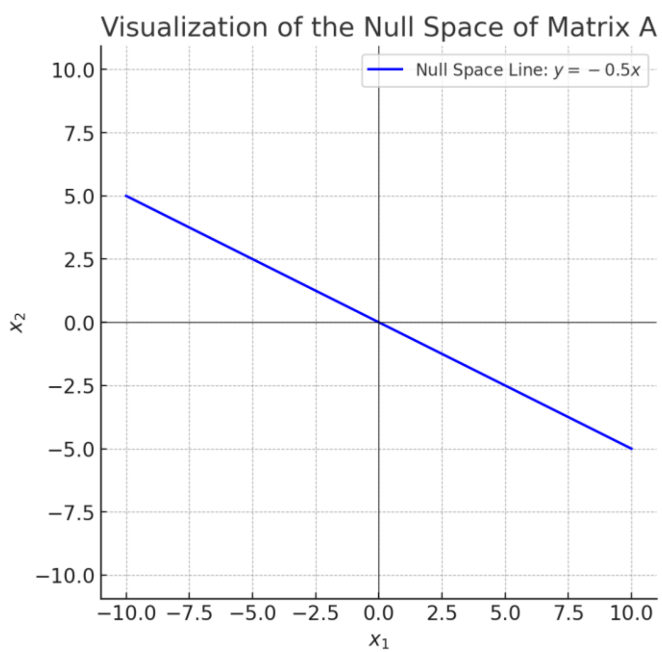
The set of vectors that land on the origin is called **null space**. For a square full-rank matrix, the only solution in the null space is 0.

If a matrix does not have full rank, the null space is not only the origin.

$$Ax = 0$$

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix} \qquad \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} x_1 + \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} x_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$x_2 = -0.5x_1$$



The line represents all vectors that, when multipied by A result in the zero vector.

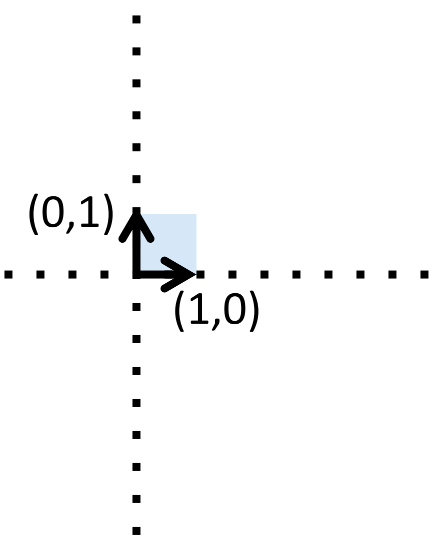
Determinant

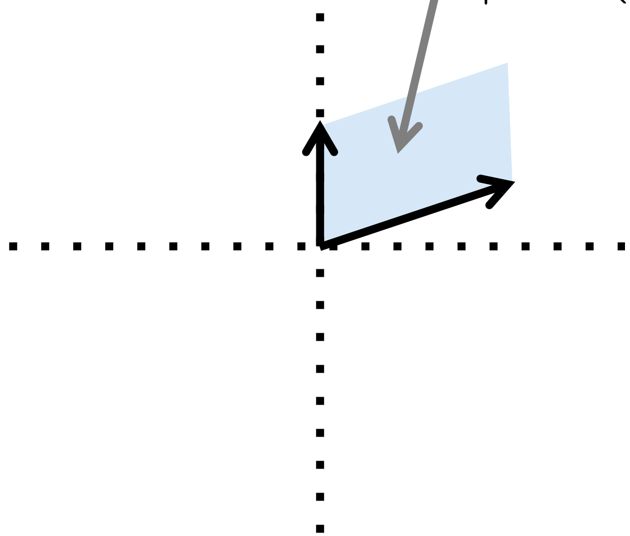
Geometrically, a determinant can be viewed as the volume **scaling** factor of the linear transformation described by the **matrix**.

$$Area = \left| det(\overleftrightarrow{M}) \right|$$

$$\begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix}$$

$$\left| det \begin{pmatrix} a & b \\ c & d \end{pmatrix} \right| = |ad - bc|$$





$$det \begin{pmatrix} 0 & 3 \\ 2 & 1 \end{pmatrix} = -6$$

A negative determinant means that the orientation is flipped.

If the determinant is zero, the volume is zero!

Symmetric and antisymmetric matrices

General (non-square) matrices *=>useful in later classes*

$A^T = A$ is a symmetric matrix

$A^T = -A$ is an antisymmetric matrix

$(AB)^T = B^T A^T$

$(A^T A)^T = A^T A$

Proposition: Any square matrix A can be uniquely decomposed into the sum of a symmetric matrix S and an antisymmetric matrix K

$$A = S + K$$

$$S = \frac{1}{2}(A + A^T), K = \frac{1}{2}(A - A^T)$$

Proof:

$$S + K = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T) = \frac{1}{2}A + \frac{1}{2}A^T + \frac{1}{2}A - \frac{1}{2}A^T = A$$

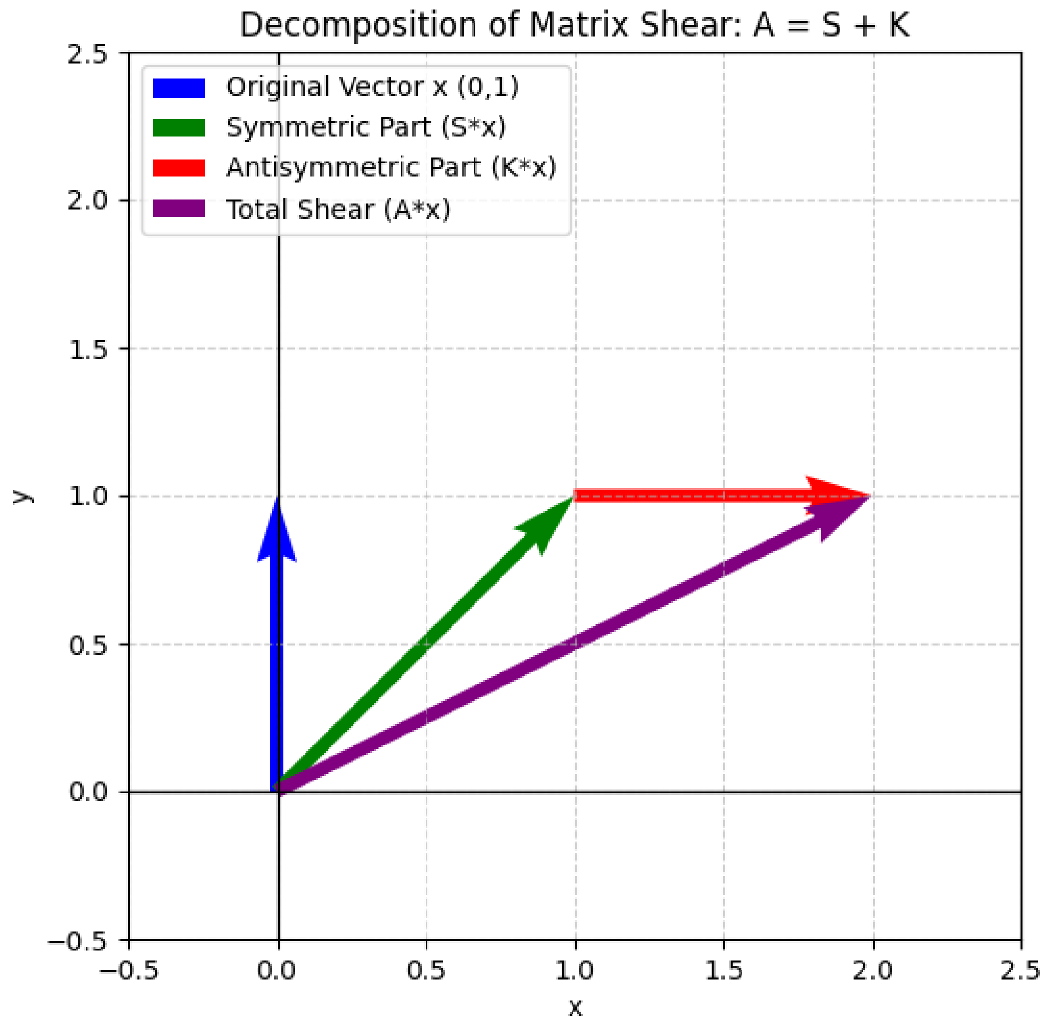
The symmetric part S stretches and squishes space along specific axes. It changes the shape. The antisymmetric part contributes rotational behavior.

Example

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$
$$S = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$
$$K = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Example

$$Ab = c = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$
$$Sb = s = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$
$$Kb = k = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$
$$s + k = c$$



Solving inhomogeneous equation systems

$Ax = b$

Possible approaches:

- Gaussian elimination method
  - transform into upper triangular matrix
  - resolve variables
- Find the inverse: $A^{-1}b = x$
- Cramer's rule (uses determinants)
- Numerous special methods …

...

Solvability of systems of linear equations

For a $m \times n$ matrix, the rank is at most $rank(A) \leq \min(m, n)$.

Theorem:
Given a $m \times n$ matrix A and an m −dimensional vector b.
The system of equations $Ax = b$ is

not solvable	if $rank(A) < rank(A|b)$.
ambiguously solvable,	if $rank(A) = rank(A|b) = r < n$.
uniquely solvable,	if $rank(A) = rank(A|b) = n$.

Theorem:
If the determinant of a matrix is $\neq 0$, the system is uniquely solvable.

The inverse of a matrix

- The inverse of a matrix A is denoted by A^{-1}.
- The inverse of a $n \times n$ matrix A is A^{-1} with $I = A^{-1}A$.
- The inverse of a $m \times n$ matrix is not defined.

Theorem 1:
A regular matrix has an inverse.

Theorem 2:
A matrix without inverse is singular.

Theorem 3:
An inverse to an $n \times n$ matrix A exists iff $|A| \neq 0$.

Generalization

- General problem (A: $m \times n$ matrix):

  $\max\ c^T x$

  $s.t.\ \ Ax = b$

  $\qquad\quad x \geq 0$

- Assume there are
  - Basic variables $\{x_{B_1}, \dots, x_{B_m}\}$
  - Non-basic variables $\{x_{N_1}, \dots, x_{N_{n-m}}\}$
- We can split the variable vector into two vectorsx_B and x_N with

  $x_B = \left[x_{B_1}\ x_{B_2}\ \dots\ x_{B_m}\right]^T$

  $x_N = \left[x_{N_1}\ x_{N_2}\ \dots\ x_{N_{n-m}}\right]^T$

- We can split the matrix A into two submatrices B and N corresponding to the basic and non-basic variables, respectively.

Method

$$Ax = b \Leftrightarrow [B\ N]\begin{bmatrix} x_B \\ x_N \end{bmatrix} = b \Leftrightarrow Bx_B + Nx_N = b$$

$$\Leftrightarrow\ B^{-1}Bx_B + B^{-1}Nx_N = B^{-1}b$$

$$\Leftrightarrow x_B = B^{-1}b - B^{-1}Nx_N$$

If $\boldsymbol{x_N} = \mathbf{0}$, this results in $\boldsymbol{x_B} = \boldsymbol{B^{-1}b}$

In general, a system with m linear equations and n unknown has $n - m$ free variables.

The above method only shows how to set the values of the remaining m variables.

Basic solutions

$$
\begin{array}{lrcrcrcr}
\max & -x_1 & - & 2x_2 & & & & \\
s.t. & 2x_1 & + & x_2 & - & 4x_3 & = & -4 \\
 & x_1 & + & 2x_2 & + & 2x_3 & = & 4 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}
$$

We again assume that the base is $\{x_1, x_3\}$.

What is B and B^{-1}?

Example

$$\begin{bmatrix} 2 & 1 & -4 \\ 1 & 2 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

$$\Leftrightarrow \left[\begin{bmatrix} 2 & -4 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right] \left[\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} \\ \begin{bmatrix} x_2 \end{bmatrix} \right] = \begin{bmatrix} 4 \\ -4 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 2 & -4 \\ 1 & 2 \end{bmatrix}}_{B} \underbrace{\begin{bmatrix} x_1 \\ x_3 \end{bmatrix}}_{x_B} + \underbrace{\begin{bmatrix} 1 \\ 2 \end{bmatrix}}_{N} \underbrace{x_2}_{x_N} = \underbrace{\begin{bmatrix} -4 \\ 4 \end{bmatrix}}_{b}$$

Observations

Given a basis, we can find the corresponding basis solution by matrix inversion.

$$x_B = B^{-1}b - B^{-1}Nx_N$$

$$\begin{aligned}
z &= c_B^T x_B + c_N^T x_N = c_B^T(B^{-1}b - B^{-1}Nx_N) + c_N^T x_N \\
&= c_B^T B^{-1}b + c_N^T x_N - c_B^T B^{-1}Nx_N \\
&= c_B^T B^{-1}b + (c_N^T - c_B^T B^{-1}N)x_N \\
&= c_B^T B^{-1}b + (c_N - (B^{-1}N)^T c_B)^T x_N
\end{aligned}$$

Conclusion

The basic solution to a basis has non-basic variables $x_N = 0$ and results in:
$x_B = B^{-1}b$

The objective value results from:
$z = c^T x = c_B^T x_B + c_N^T x_N$, with
$c_B^T = [c_{B_1}\ c_{B_2}\ \dots\ c_{B_m}]$ and $c_N^T = [c_{N_1}\ c_{N_2}\ \dots\ c_{N_{n-m}}]$

Given the basic solution with $x_B = B^{-1}b$ and $x_N = 0$, this results in a objective value of
$z = c_B^T B^{-1} b$ for the basis.

Observation

The 2 equations of the original system

$$z - c^T x = 0$$
$$Ax = b$$

can be formulated as:

$$z - (c_N - (B^{-1}N)^T c_B)^T x_N = c_B^T B^{-1} b$$

$$x_B + B^{-1} N x_N = B^{-1} b$$

Standard form in terms of the basis!

Simplex in matrix notation

- Current non-basic coefficients: $N' = B^{-1}N$

- Entering basic variable:
  – is determined by $c_N' = -(c_N - N'^T c_B)$

- Exiting basic variable:
  – We need to know the column i of the entering basic variable.
  – The exiting BV is determined by N_i' and $b' = x_B = B^{-1}b$.

- If B^{-1} is known, these steps can be performed quickly.

Comparison to Simplex tableau

<table>
<tr><th rowspan="2">Basic Variable</th><th colspan="7">Coefficient</th><th rowspan="2">Solution</th></tr>
<tr><th>z</th><th>x_{B_1}</th><th>$\dots$</th><th>x_{B_m}</th><th>x_{N_1}</th><th>$\dots$</th><th>$x_{N_{n-m}}$</th></tr>
<tr><td>z</td><td>1</td><td>0</td><td></td><td>0</td><td colspan="3">$c_N'^T = -(c_N - N'^T c_B)^T$</td><td>$c_B^T B^{-1} b$</td></tr>
<tr><td>x_{B_1}</td><td></td><td>1</td><td></td><td></td><td colspan="3" rowspan="3">$N' = B^{-1} N$</td><td rowspan="3">$b' = B^{-1} b$</td></tr>
<tr><td>$\dots$</td><td></td><td></td><td>$\dots$</td><td></td></tr>
<tr><td>x_{B_m}</td><td></td><td></td><td></td><td>1</td></tr>
</table>

Comparison to Simplex tableau

$$\begin{array}{lcccccccc}
\text{max} & -x_1 & - & 2x_2 & & & & \\
\text{s.t.} & 2x_1 & + & x_2 & - & 4x_3 & = & -4 \\
 & x_1 & + & 2x_2 & + & 2x_3 & = & 4 \\
 & & & & & x_1, x_2, x_3 & \geq & 0
\end{array}$$

Start tableau from last time (after using the Big M method):

Basic variable	Coefficient				Solu-tion
	z	x_1	x_3	x_2	
z	1			0.75	-1
x_1		1		1.25	1
x_3			1	0.375	1.5

$c'_N = -(c_N - (B^{-1}N)^T c_B)$
$= -(-2 - (-1.25)) = 0.75$

$Z' = c_B^T B^{-1} b$

$$B^{-1}N \Rightarrow \begin{pmatrix} 0.25 & 0.5 \\ -0.125 & 0.25 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 1.25 \\ 0.375 \end{pmatrix}$$

$$B^{-1}b \Rightarrow \begin{pmatrix} 0.25 & 0.5 \\ -0.125 & 0.25 \end{pmatrix} \begin{pmatrix} -4 \\ 4 \end{pmatrix} = \begin{pmatrix} 1 \\ 1.5 \end{pmatrix}$$

Comparison to Simplex tableau

Basic variable	Coefficient				Solution
	z	x_1	x_3	x_2	
z	1			0.75	-1
x_1		1		1.25	1
x_3			1	0.375	1.5

<table>
<tr><th rowspan="2">Basic variable</th><th colspan="7">Coefficient</th><th rowspan="2">Solution</th></tr>
<tr><th>z</th><th>x_{B_1}</th><th>$\dots$</th><th>x_{B_m}</th><th>x_{N_1}</th><th>$\dots$</th><th>$x_{N_{n-m}}$</th></tr>
<tr><td>z</td><td>1</td><td>0</td><td></td><td>0</td><td colspan="3">$-(c_N - (B^{-1}N)^T c_B)^T$</td><td>$c_B^T B^{-1} b$</td></tr>
<tr><td>x_{B_1}</td><td></td><td>1</td><td></td><td></td><td colspan="3" rowspan="3">$B^{-1}N$</td><td rowspan="3">$B^{-1}b$</td></tr>
<tr><td>$\dots$</td><td></td><td></td><td>$\dots$</td><td></td></tr>
<tr><td>x_{B_m}</td><td></td><td></td><td></td><td>1</td></tr>
</table>

Observations

A basic solution is completely determined by the partial matrix B, which consists of the basic variables.

- The feasibility of a solution is determined by the value of the matrix $x_B = B^{-1}b$.
- The optimality of the solution is determined by the value of the vector

  $c'_N = -(c_N - N'^T c_B)$.

We can call it an optimal basis instead of an optimal solution.

Revised Simplex

Determining a new basic feasible solution is not done by elementary row operations as in the regular Simplex method, but with the help of matrix operations.

Advantages:

- smaller memory requirement
- smaller number of arithmetic operations
- rounding errors have less effect, because this method uses initial values.

Simplex in matrix notation

- Find an initial set of basic variables
- Repeat {
  - Determine B^{-1}
  - Calculate $N' = B^{-1}N$
  - Find the entering basic variable by calculating $c_N' = -(c_N - N'^T c_B)$
  - Use N' to find the exiting basic variable.
  - Basis swap, compute B, N, c_B, c_N

} until optimality detected, or the problem is unbounded.

Revised Simplex 1/6

$$\begin{aligned} &max\ \ 3000x_1 + 2000x_2 \\ &s.t. \qquad 5x_1 + \quad 2x_2 + x_3 \qquad = 180 \\ &\qquad\qquad 3x_1 + \quad 3x_2 \qquad + x_4 = 135 \\ &\qquad\qquad\qquad\qquad x_1, x_2, x_3, x_4 \geq \quad 0 \end{aligned}$$

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	-3000	-2000			0
x_3		5	2	1		180
x_4		3	3		1	135

New basic variable: x_1, x_4
New non-basic variable: x_2, x_3

Revised Simplex 2/6

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1	-3000	-2000			0
x_3		5	2	1		180
x_4		3	3		1	135

New basic variable: x_1, x_4
New non-basic variable: x_2, x_3

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1		-800	600		108000
x_1		1	2/5	1/5		36
x_4			9/5	-3/5	1	27

Revised Simplex 3/6

$$\begin{aligned} max\ \ & 3000x_1 + 2000x_2 \\ s.t.\ \ & 5x_1 + 2x_2 + x_3 = 180 \\ & 3x_1 + 3x_2 + x_4 = 135 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

$$z - (c_N - (B^{-1}N)^T c_B)^T x_N = c_B^T B^{-1} b$$

$$x_B + (B^{-1}N)^T x_N = B^{-1} b$$

$A = \begin{pmatrix} 5 & 2 & 1 & 0 \\ 3 & 3 & 0 & 1 \end{pmatrix}$ $\quad b = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$ $\quad c^T = (3000, 2000, 0, 0)$

Initial basis x_3, x_4 Non-basic variables: x_1, x_2

$B = B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $\quad b' = x_B = B^{-1}b = \begin{pmatrix} x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$

$N = \begin{pmatrix} 5 & 2 \\ 3 & 3 \end{pmatrix}$ $\quad c_N^T = (3000, 2000)$ $\quad c_B^T = (0, 0)$ $\quad z = c_B^T B^{-1} b = 0$

Search entering basic variable:

$c_N'^T = -(c_N - (B^{-1}N)^T c_B)^T = (\boxed{-3000,}\ -2000)$

New basis: x_1, x_4
New NBV: x_2, x_3

Search exiting basic variable:

$N' = B^{-1}N = \begin{pmatrix} \boxed{5} & 2 \\ 3 & 3 \end{pmatrix}$

$$\boxed{\begin{aligned} 180/5 &= 36 \\ 135/3 &= 45 \end{aligned}}$$

Revised Simplex 4/6

$$A = \begin{pmatrix} 5 & 2 & 1 & 0 \\ 3 & 3 & 0 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$$

$$c^T = (3000, 2000, 0, 0)$$

$$z - (c_N - (B^{-1}N)^T c_B)^T x_N = c_B^T B^{-1} b$$

$$x_B + (B^{-1}N)^T x_N = B^{-1} b$$

Basis x_1, x_4

Non-basic variables x_2, x_3

$$B = \begin{pmatrix} 5 & 0 \\ 3 & 1 \end{pmatrix} \qquad N = \begin{pmatrix} 2 & 1 \\ 3 & 0 \end{pmatrix} \qquad b' = x_B = B^{-1} b = \begin{pmatrix} x_1 \\ x_4 \end{pmatrix} = \begin{pmatrix} 36 \\ 27 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 1/5 & 0 \\ -3/5 & 1 \end{pmatrix}$$

$$c_N^T = (2000, 0) \qquad c_B^T = (3000, 0)$$

$$z = c_B^T B^{-1} b = (3000{,}0) \cdot \begin{pmatrix} 1/5 & 0 \\ -3/5 & 1 \end{pmatrix} \cdot \begin{pmatrix} 180 \\ 135 \end{pmatrix} = 108000$$

Search entering basic variable:

$$c_N'^T = -(c_N - (B^{-1}N)^T c_B)^T = (\boxed{-800}, 600)$$

Search exiting basic variable:

$$N' = B^{-1} N = \begin{pmatrix} 2/5 & 1/5 \\ \boxed{9/5} & -3/5 \end{pmatrix}$$

$$\begin{array}{l} 36/(2/5) = 90 \\ 27/(9/5) = 15 \end{array}$$

New Basis: x_1, x_2
New NBV: x_3, x_4

Revised Simplex 5/6

$$A = \begin{pmatrix} 5 & 2 & 1 & 0 \\ 3 & 3 & 0 & 1 \end{pmatrix} \qquad b = \begin{pmatrix} 180 \\ 135 \end{pmatrix}$$

$$c^T = (3000, 2000, 0, 0)$$

$$z - (c_N - (B^{-1}N)^T c_B)^T x_N = c_B^T B^{-1} b$$

$$x_B + (B^{-1}N)^T x_N = B^{-1} b$$

Basis x_1, x_2 Non-basic variables x_3, x_4

$$B = \begin{pmatrix} 5 & 2 \\ 3 & 3 \end{pmatrix} \qquad N = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad b' = x_B = B^{-1} b = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 30 \\ 15 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 1/3 & -2/9 \\ -1/3 & 5/9 \end{pmatrix}$$

$$c_N^T = (0, 0) \qquad c_B^T = (3000, 2000)$$

$$z = c_B^T B^{-1} b = (3000{,}2000) \cdot B^{-1} \cdot \begin{pmatrix} 180 \\ 135 \end{pmatrix} = 120000$$

Search entering basic variable:

$$c_N'^T = -(c_N - (B^{-1}N)^T c_B)^T = \left(\frac{1000}{3}, \frac{4000}{9}\right)$$

→ Optimal

Revised Simplex 6/6

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1		-800	600		108000
x_1		1	2/5	1/5		36
x_4			9/5	-3/5	1	27

New basic variables: x_1, x_4
New non-basic variables: x_2, x_3

Basic variable	Coefficient					Solution
	z	x_1	x_2	x_3	x_4	
z	1			1000/3	4000/9	120000
x_1		1		1/3	-2/9	30
x_2			1	-1/3	5/9	15

Summary

Revised Simplex

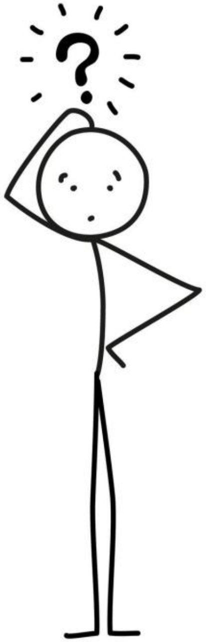
- Simple way to calculate values and the corresponding solution only by knowing the basic variables

Allows simple sensitivity analysis

- integration of new variables
- re-optimization
- Does the optimal solution change when the parameters change?

Please study the formulas that we derived for the next class.

What is the difference between the previous algebraic Simplex and the revised Simplex?



Today you will learn ...

- how to formulate the Simplex algorithm in matrix notation and how to run the **revised Simplex algorithm**.
  - This will be the central notation we use in the next lectures.

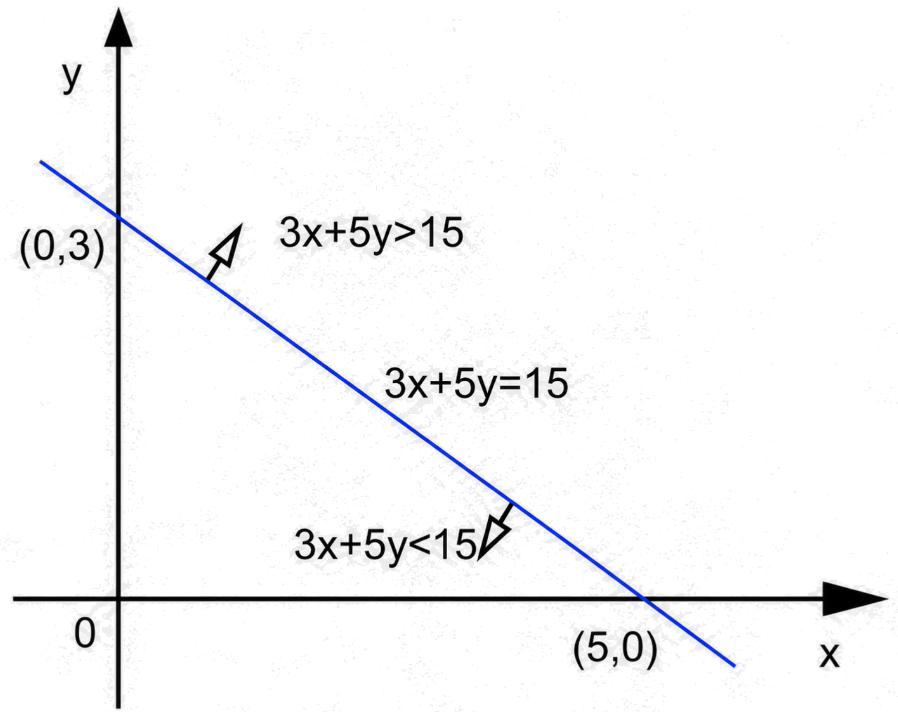
- Apply linear programming for **classification problems**.

- Understand **vector optimization** and how you deal with conflicting objectives.

If you haven‘t done so, please exercise to understand the Simplex algorithm.

Hyperplanes and half-spaces

Hyperplane: $H = \{x : \langle w, x \rangle = b\}$
Half-space: $H^+ = \{x : \langle w, x \rangle \leq b\}, H^- = \{x : \langle w, x \rangle \geq b\}$



Application to data analysis

The connection to Machine Learning:

We learned that an LP constraint $w^T x \leq b$ defines a **half-space**.

The intersection of constraints forms a polytope.

In **Machine Learning**, we use these same hyperplanes to separate data classes.

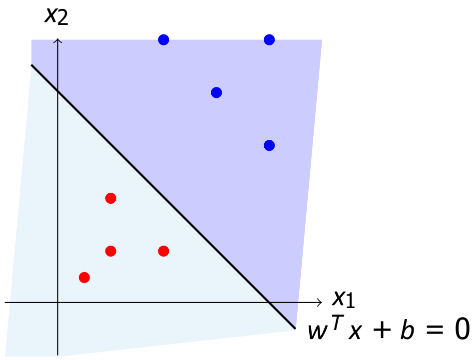
The goal:

Given a dataset of **Blue** and **Red** points.

Find a linear inequality (hyperplane) that separates them.

This is the feasibility version underlying linear classification. Support Vector Machines (SVM)

Add a margin-maximizing objective.



Modeling linear classification as an LP

Let our dataset be n points $x_i \in \mathbb{R}^d$ with labels $y_i \in \{+1, -1\}$. We want to find a weight vector w and bias b such that:

$$(x_1, y_1) = ((2,2), +1),$$
$$(x_2, y_2) = ((3,1), +1),$$
$$(x_3, y_3) = ((1,3), +1),$$
$$(x_4, y_4) = ((1,1), -1),$$

$w^T x_i + b > 0$ if $y_i = +1$ (Blue)
$w^T x_i + b < 0$ if $y_i = -1$ (Red)

To enforce a strict margin, we normalize the requirement:

$$y_i (w^T x_i + b) \geq 1 \ \forall i = 1, \ldots, n$$

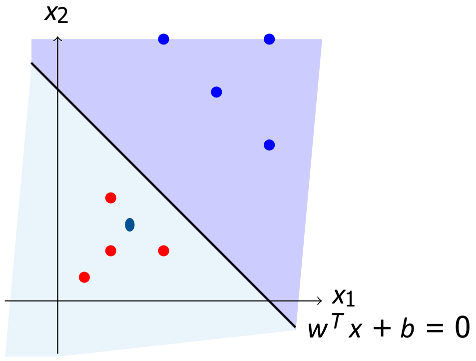
This is just a system of linear inequalities!

We can feed this into the Simplex algorithm to find valid w, b.

Data is noisy

Problem: Real-world data is rarely perfectly separable with a linear hyperplane.

A "hard" Simplex solver would return: **Infeasible**.



Solution: Allow mistakes, but minimize them.

We introduce **slack variables** $\xi_i \geq 0 \ (X_i)$.

ξ_i measures the violation of the margin constraint.

The classification LP

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{n} \xi_i \\
\text{s.t.} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i \quad \forall i \\
& \xi_i \geq 0 \quad \forall i
\end{aligned}
$$

We want to satisfy constraints as much as possible, while minimizing total sum of errors.

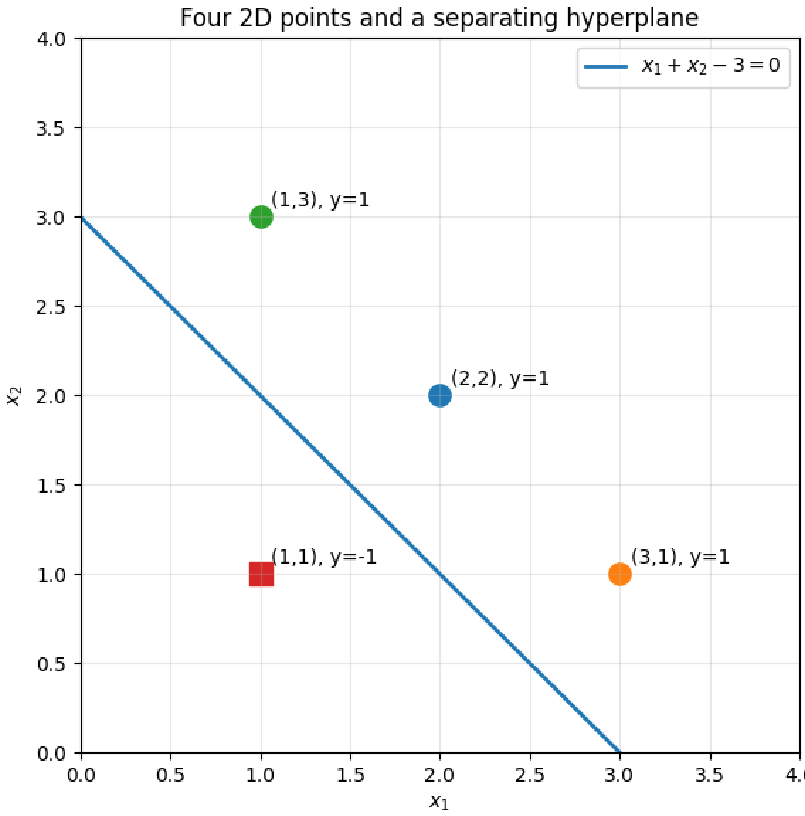
A numerical example

$$
\begin{aligned}
\min \quad & \sum_{i=1}^{n} \xi_i \\
\text{s.t.} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i \quad \forall i \\
& \xi_i \geq 0 \quad \forall i
\end{aligned}
$$

$$(x_1, y_1) = ((2,2), +1),$$
$$(x_2, y_2) = ((3,1), +1),$$
$$(x_3, y_3) = ((1,3), +1),$$
$$(x_4, y_4) = ((1,1), -1),$$

$$\min_{w_1, w_2, b, \xi_1, \xi_2, \xi_3, \xi_4} \quad \xi_1 + \xi_2 + \xi_3 + \xi_4$$

$$(+1)\big(w_1 \cdot 2 + w_2 \cdot 2 + b\big) \geq 1 - \xi_1,$$
$$(+1)\big(w_1 \cdot 3 + w_2 \cdot 1 + b\big) \geq 1 - \xi_2,$$
$$(+1)\big(w_1 \cdot 1 + w_2 \cdot 3 + b\big) \geq 1 - \xi_3,$$
$$(-1)\big(w_1 \cdot 1 + w_2 \cdot 1 + b\big) \geq 1 - \xi_4,$$
$$\xi_1 \geq 0, \qquad \xi_2 \geq 0, \qquad \xi_3 \geq 0, \qquad \xi_4 \geq 0$$



$$\boxed{w = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad b = -3.}$$

Separating line:

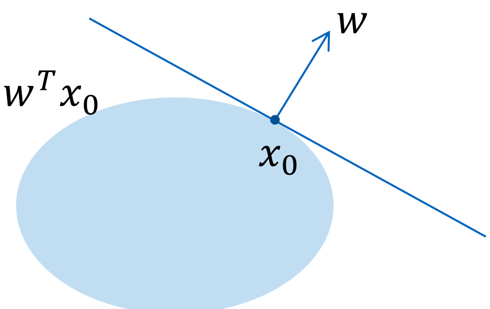
$$x_1 + x_2 - 3 = 0$$

Supporting hyperplane

Given a hyperplane $H = \{x \in \mathbb{R}^n | w^T x = b\}$, we say that the hyperplane H passes through a point x_0 when $x_0 \in H$ which is equivalent to $w^T x_0 = b$.

One of the closed halfspaces determined by the hyperplane contains X when either $w^T x \leq b$ for all $x \in X$, or $w^T x \geq b$ for all $x \in X$

$$\sup_{z \in X} w^T z \leq w^T x_0$$



Supporting hyperplane theorem: Let $X \subseteq \mathbb{R}^n$ be a nonempty convex set and x_0 be at the boundary of this set. Then, there exists a hyperplane passing through x_0 and containing the set X in one of its halfspaces. This means, there is a vector $w \in \mathbb{R}^n, w \neq 0$, such that $\sup_{z \in X} w^T z = w^T x_0$. This is a *supporting hyperplane*.

Multi-objective optimization

LP model up until now:

– optimization **of a single objective** (e.g., maximize profit)
– under constraints (e.g., capacities).

In some cases, however, there are **several competing objectives** to be considered, e.g.:

– profit as high as possible
– material consumption as low as possible (so-called "soft" constraint)
– with more (hard) constraints

This is also referred to as **vector optimization.**

Solutions with multiple objectives

Methods without preferences (e.g. $\min\|f(x) - z^{ideal}\|$, s.t. $x \in X$)

Selection with predefined preferences of the decision maker (**a priori methods**):

- If there is a dominant objective, other objectives can possibly be mapped via constraints that determine a minimum target value
- Lexicographic order of goals
- Linear scalarization: the objective functions can be weighted and added together $\sum w_i f_i(x)$
  - The determination of the weights requires care. There are numerous procedures to elicit weights as they can have substantial effect.
- Goal programming

Goal Programming (GP)

GP seeks the solution that achieves all goals as much as possible.

Example: A company specifies targets for the number of customer contacts made by sales personnel per month:

- Target 1: 200 existing customers
- Target 2: 120 new customers

Each contact with existing customers takes an average of 2 hours.

Each new customer contact requires an average of 3 hours.

The company has a maximum of 640 hours available with the current sales staff per month.

– $200(2) + 120(3) = 760 > 640$

Goal constraints

Decision variables:

x_1 = number of existing customers to be contacted

x_2 = number of new customers to be contacted

Goal 1: $x_1 - d_1^+ + d_1^- = 200$

d_1^+ = number of contacts to the existing customers above 200

d_1^- = number of contacts to the existing customers below 200

Goal 2: $x_2 - d_2^+ + d_2^- = 120$

d_2^+ = number of new customer contacts above 120

d_1^- = number of new customer contacts below 120

GP formulation

$$\min \; d_1^- + d_2^-$$

$$
\begin{aligned}
s.t. \quad & x_1 - d_1^+ + d_1^- = 200 && \text{(deviation from existing customer target)}\\
& x_2 - d_2^+ + d_2^- = 120 && \text{(deviation from new customer target)}\\
& 2x_1 + 3x_2 \le 640 && \text{(available hours)}\\
& x_1, x_2, d_1^+, d_1^-, d_2^+, d_2^- \ge 0
\end{aligned}
$$

Optimal solution:

$x_1 = 200,$ $x_2 = 80,$ $d_2^- = 40$

Depending on the nature of the objective, either d_i^-, d_i^+ or both variables appear in the objective function.

Weighted GP formulation

What happens when the company values new customer contacts more than contacts with existing customers?

For example, if goal 2 is two times as important as goal 1 (or yields two times more):

- objective function: min $d_1^- + 2d_2^-$
- same constraints as above
- new solution: $x_1 = 140, \; x_2 = 120, \; d_1^- = 60$